

There are a total of seven parts in problem 1. You have to solve all of them.

Problem 1 (CO1): Deterministic Finite Automata (20 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{w \text{ contains at least three 1s}\}$$

$$L_2 = \{w \text{ starts and ends with different symbols}\}$$

$$L_3 = \{w \text{ doesn't contain 01 as a subsequence}\}$$

$$L_4 = \{w \text{ ends with at least one 0}\}$$

Now solve the following problems.

- (a) **Design** a DFA for the language L_1 . (4 points)
- (b) **Design** a DFA for the language L_2 . (4 points)
- (c) **Design** a DFA for the language L_3 . (4 points)
- (d) **Write** all four-letter strings in $L_1 \cap L_4$. (1 point)
- (e) **Design** a DFA for the language $L_1 \cap L_4$. (3 points)
- (f) **Write** all four-letter strings in $L_2 \cap L_4$. (1 point)
- (g) **Design** a DFA for the language $L_2 \cap L_4$. (3 points)

Answer to the Q. NO-01(a)

Let, $\Sigma = \{0, 1\}$

Given, $L_1 = \{w \text{ contains at least three 1s}\}$

Now,

For DFA for given L_1 ,

$q_0 = 0$ ones

$q_1 = 1$ one

$q_2 = 2$ ones

$q_3 = 3$ or more ones.

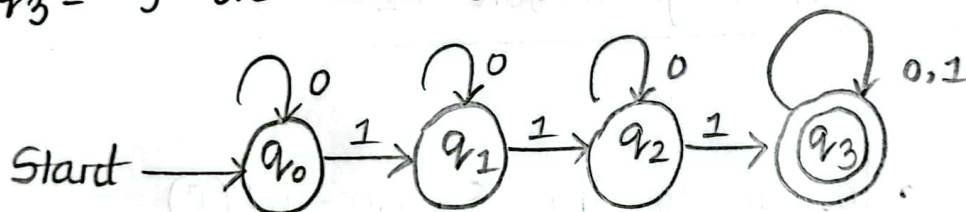


Fig: DFA diagram for L_1

Answer to the Q. NO - 01(b)

Given,

$L_2 = \{ w \text{ starts and ends with different symbols} \}$

Now,

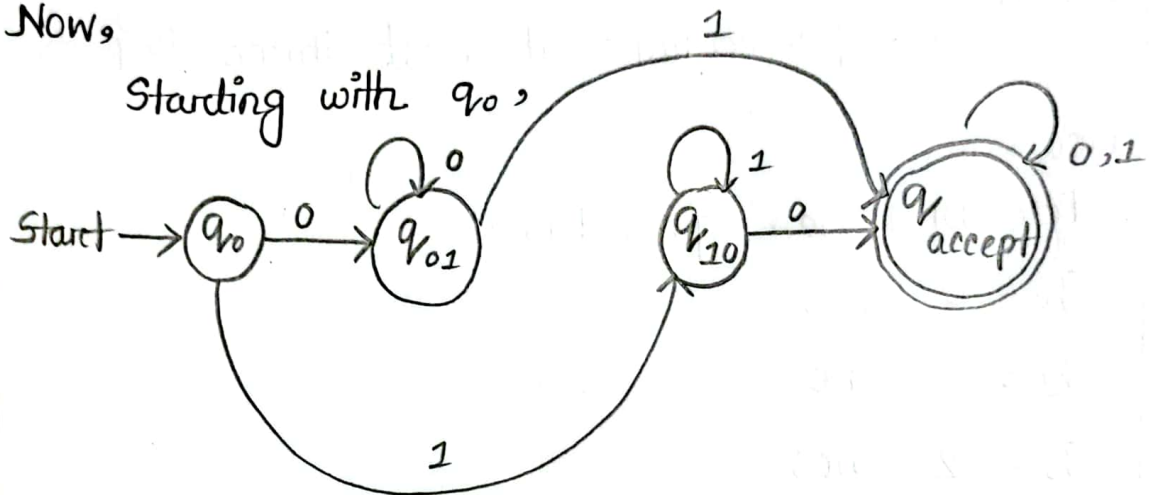


Fig : DFA diagram for L_2

Answer to the Q. NO - 01(c)

Given, $L_3 = \{ w \text{ doesn't contain } 01 \text{ as a subsequence} \}$

Now,

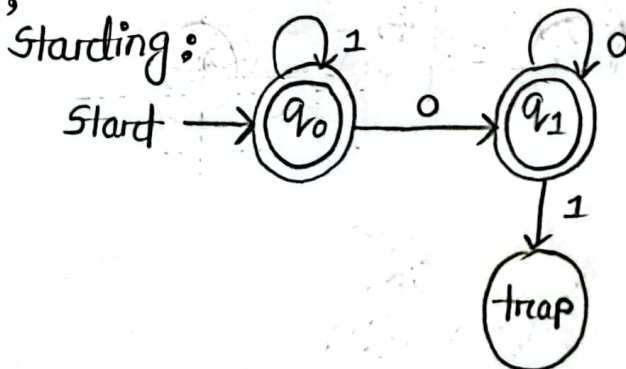


Fig : DFA diagram for L_3

[P.T.O.]

Answer to the Q. NO- 01 (d)

Given,

$$L_1 = \{w \text{ contains at least three 1s}\}$$

$$L_4 = \{w \text{ ends with at least one 0}\}$$

For $L_1 \cap L_4$,

The strings:

(i) 1110

(ii) 1100

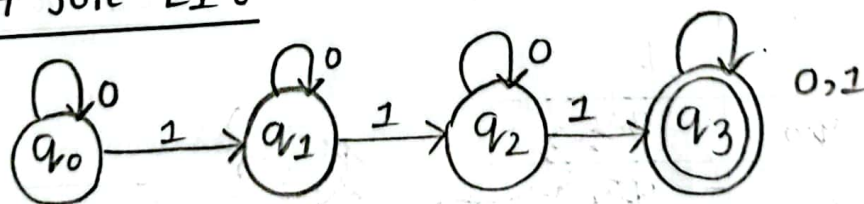
(iii) 1010

(iv) 0110

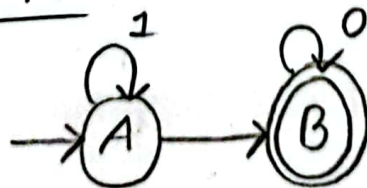
Answer to the Q. NO- 01 (e)

For $L_1 \cap L_4$,

The DFA for L_1 :



The DFA for L_4 :



[P.T.O.]

Here,

$$M_1 = \{q_0, q_1, q_2, q_3\}$$

$$M_2 = \{A, B\}$$

$$\therefore M = M_1 \times M_2$$

$$= \{ (q_0A), (q_1A), (q_2A), (q_3A), (q_0B), (q_1B), (q_2B), (q_3B) \}$$

Here,

$$\text{Start State} = \{q_0A\}$$

$$\text{set of final states, } L_1 = \{q_3\}$$

$$\text{and } L_2 = \{B\}$$

$$\text{Final State} = \{q_3B\}$$

Now,

Transition Table for L_1 :

	0	1
$\rightarrow q_0$	q_0	q_1
q_1	q_1	q_2
q_2	q_2	q_3
$\otimes q_3$	q_3	q_3

Transition Table for L_2 :

	0	1
A	B	A
$\otimes$ B	B	A

[P.T.O.]

Transition of the new DFA
(Input symbol 0 and 1)

$$\delta(q_0A, 0) = q_0A$$

$$\delta(q_1A, 0) = q_1B$$

$$\delta(q_2A, 0) = q_2B$$

$$\delta(q_3A, 0) = q_3B$$

$$\delta(q_0B, 0) = q_0B$$

$$\delta(q_1B, 0) = q_1B$$

$$\delta(q_2B, 0) = q_2B$$

$$\delta(q_3B, 0) = q_3B$$

$$\delta(q_0A, 1) = q_1A$$

$$\delta(q_1A, 1) = q_2A$$

$$\delta(q_2A, 1) = q_3A$$

$$\delta(q_3A, 1) = q_3A$$

$$\delta(q_0B, 1) = q_1A$$

$$\delta(q_1B, 1) = q_2A$$

$$\delta(q_2B, 1) = q_3A$$

$$\delta(q_3B, 1) = q_3A$$

DFA:

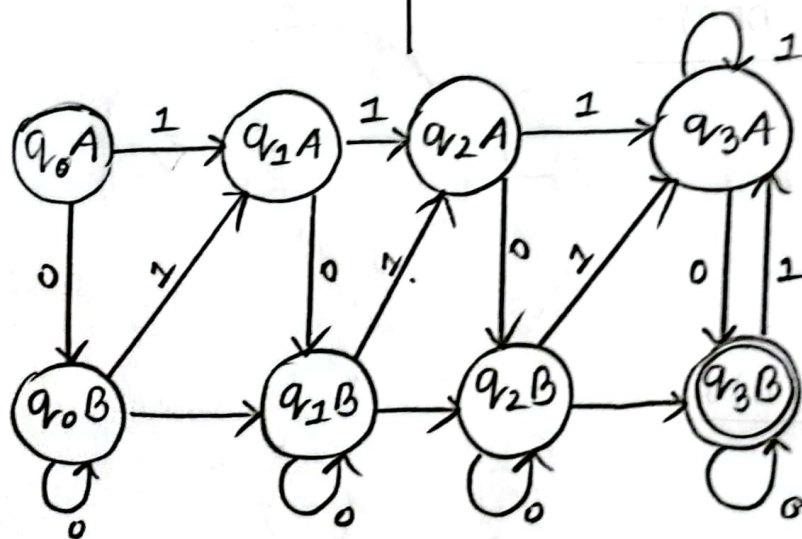


Fig: State diagram of DFA for $L_1 \cap L_4$

[P.T.O.]

Answer to the Q. NO - 01 (f)

Given,

$L_2 = \{w \text{ starts and ends with different symbols}\}$

$L_4 = \{w \text{ ends with at least one } 0\}$

Now, all four-letter strings for $L_2 \cap L_4$ will be :

(i) 1 0 0 0

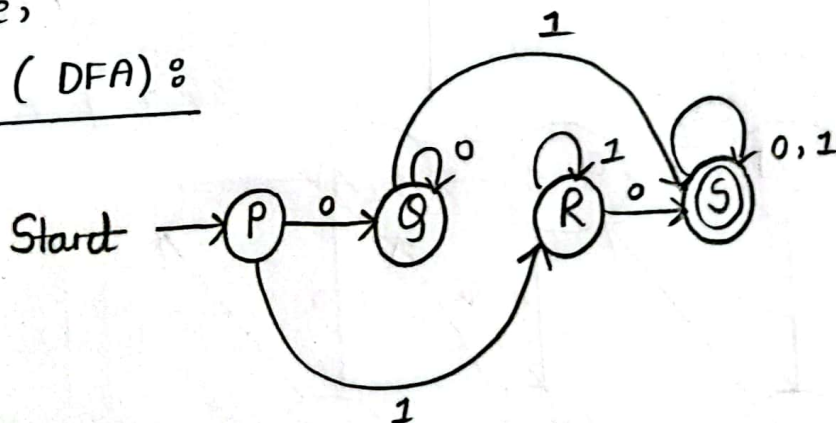
(ii) 0 1 0 0

(iii) 0 1 0 1

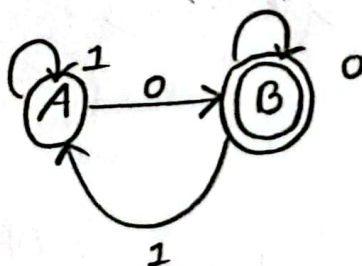
Answer to the Q. NO - 01 (g)

Here,

L_2 (DFA) :



L_4 (DFA) :



[P.T.O.]

here,

$$M_1 = \{P, Q, R, S\}$$

$$M_2 = \{A, B\}$$

$$\therefore M = M_1 \times M_2$$

$$= \{(PA), (QA), (RA), (SA), (PB), (QB), (RB), (SB)\}$$
$$= \{(AP), (AQ), (AR), (AS), (BP), (BQ), (BR), (BS)\}$$

Now, Starting with state = AP
Final or ending state = BS

Transition Table for L_2 :

	0	1
P	Q	R
Q	Q	S
R	S	R
S	S	S

Transition Table for L_4 :

	0	1
A	B	A
B	B	A

[P.T.O.]

Transition for DFA (new) $\rightarrow L_2 \cap L_4$

$$\delta(AP, 0) = BQ$$

$$\delta(AQ, 0) = BQ$$

$$\delta(AR, 0) = BS$$

$$\delta(AS, 0) = BS$$

$$\delta(BP, 0) = BQ$$

$$\delta(BQ, 0) = BQ$$

$$\delta(BR, 0) = BS$$

$$\delta(BS, 0) = BS$$

$$\delta(AP, 1) = AR$$

$$\delta(AQ, 1) = AS$$

$$\delta(AR, 1) = AR$$

$$\delta(AS, 1) = AS$$

$$\delta(BP, 1) = AR$$

$$\delta(BQ, 1) = AS$$

$$\delta(BR, 1) = AR$$

$$\delta(BS, 1) = AS$$

$\therefore$ DFA for $L_2 \cap L_4$:

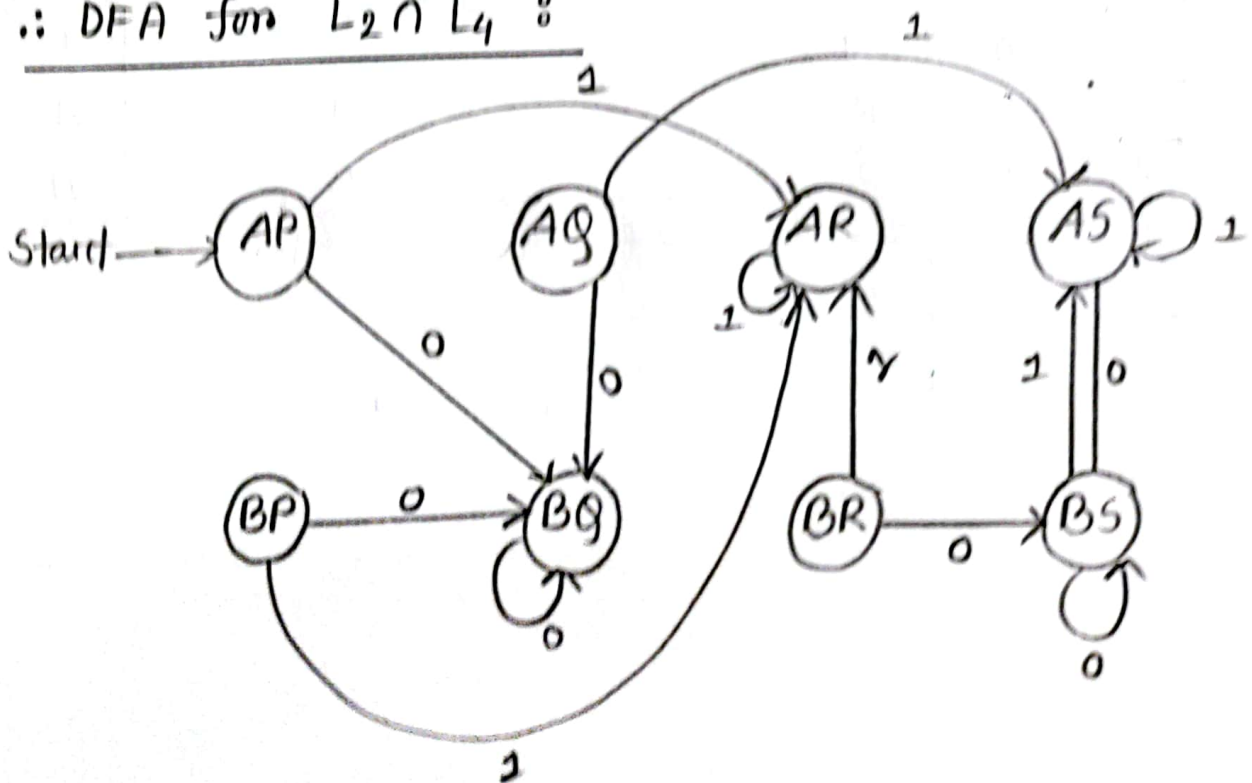


Fig 1 DFA diagram for $L_2 \cap L_4$

There are a total of five problems. You have to solve all of them.

Problem 1 (CO5): Nonregular Language (25 points)

Use the pumping lemma to **demonstrate** that L_1, L_2, L_3, L_4 and L_5 is not regular.

- (a) $L_1 = \{ww \mid w \in \{0,1\}^*\}$ (5 points)
- (b) $L_2 = \{w \in \{0,1\}^* : 0^x 1^y 0^z \text{ where } z > x + y \text{ and } x, y \geq 0\}$ (5 points)
- (c) $L_3 = \{w \in \{0,1\}^* : w \text{ is a palindrome}\}$ (5 points)
- (d) $L_4 = \{w \in \{a,b\}^* : \text{numbers of a in } w \text{ is a prime number}\}$ (5 points)
- (e) $L_5 = \{w \in \{0\}^* : 0^{3^n} \text{ where } n \geq 0\}$ (5 points)



Assignment - 02

Submitted by : TASNIM RAHMAN MOUMITA

ID : 22301689

Course Title : Automata & Computability

Course Code : CSE331

Section : 09

Date of submission : 27. 12. 2024

Semester : Fall 2024

There are a total of five problems. You have to solve all of them.

Problem 1 (CO5): Nonregular Language (25 points)

Use the pumping lemma to **demonstrate** that L_1, L_2, L_3, L_4 and L_5 is not regular.

- (a) $L_1 = \{ww \mid w \in \{0,1\}^*\}$ (5 points)
- (b) $L_2 = \{w \in \{0,1\}^* : 0^x 1^y 0^z \text{ where } z > x + y \text{ and } x, y \geq 0\}$ (5 points)
- (c) $L_3 = \{w \in \{0,1\}^* : w \text{ is a palindrome}\}$ (5 points)
- (d) $L_4 = \{w \in \{a,b\}^* : \text{numbers of a in } w \text{ is a prime number}\}$ (5 points)
- (e) $L_5 = \{w \in \{0\}^* : 0^{3^n} \text{ where } n \geq 0\}$ (5 points)



Answer to the Q. NO - 01(a)

Given,

$$L_1 = \{ww \mid w \in \{0,1\}^*\}$$

Step-1: Assumption: let us assume,
 L_1 is regular.

Step-2: Pumping Length: let,
 $p =$ pumping length

Step-3: Choose :

for string, $S_1 = 0101 \in L_1$

here, $S_1 = 0^p 1^p 0^p 1^p \in L_1$

Dividing S_1 into xyz ;

such that $|xy| \leq p$ &
 $|y| > 0$

here, $|xy| \leq p$

$xy \rightarrow$ consists of only for 0's.

Now, pumping y ,

$$xyy'z = 0^{p+k} 1^p 0^p 1^p$$

It is not in L_1 , because it is not in form ww .
 $\therefore$ "There is a contradiction."

Step-4: Conclusion: Therefore, L_1 is non-regular.

[proved]

[P.T.O.]

Answers to the Q. NO - 01(b)

Given,

$$L_2 = \{w \in \{0,1\}^* : 0^x 1^y 0^z \text{ where } z > x+y \text{ and } x, y \geq 0\}$$

Step-1: Assumption:

Let us assume,

L_2 is regular.

Step-2: Pumping length: Let, p = pumping length.

for a string, $S_2 = 0^p 1^p 0^{2p+1}$

Step-3: Choose:

Here,

$S_2 \Rightarrow$ splitting into xyz ,

Here,

$$x = p$$

$$y = p$$

$$\therefore z = 2p+1;$$

It satisfies $\Rightarrow$

$$z > x+y$$

Now,

$$S_2' = xy^2z$$

$$\therefore S_2' = 0^{p+|y|} 1^p 0^{2p+1}$$

Here,

x increases,

where,

$$|xy| \leq p$$

$$|y| > 0$$

$\therefore y$ consists of only 0's.

but, $z > x+y$; [no longer

holds; as $x+y$ increases beyond z .]

Hence, There is a contradiction.

Step-4: Conclusion: Therefore, L_2 is non-regular.

[proved]

[P.T.O.]

Answer to the Q. NO-01(c)

Given,

$$L_3 = \{ \omega \in \{0,1\}^* : \omega \text{ is a palindrome} \}$$

Step-1: Assumption: let us assume,
 L_3 is regular.

Step-2: Pumping length: let,
 $p =$ pumping length.

Step-3: Choose:
for string, $S_3 = 0^p \cdot 10^p \in L_3$

Splitting S_3 into xyz ,

$$S_3 = 0^x 10^z;$$

And,

pumping y ,

$$xyy^kz = 0^{(p+k)} 10^p$$

where, $x = p-k$

$$z = p$$

$$0 < k \leq p$$

S_3 consists of
 p 0s, followed by
a single 1 then
another p 0s.

which creates
symmetry.

Here,

Increases number of 0s before the 1, which
breaks the symmetry of palindrome.

$\therefore$ There is a contradiction.

Step-4: Conclusion: Therefore, L_3 is non-regular.

[proved]

[P.T.O.]

Answer to the Q. NO-01(d)

Given,

$$L_4 = \{w \in \{a,b\}^* : \text{numbers of } a \text{ in } w \text{ is a prime number}\}$$

Step-1: Assumption:

let us assume,

L_4 is regular.

Step-2: Pumping Length:

let, $p =$ pumping length.

Step-3: Choose:

for string, $S_4 = a^p \in L_4$

Now,

Splitting S_4 into xyz ,

here, $p = \text{prime}$.

let,

$$y = a^m ; m > 0$$

Where,

$$|xy| \leq p \&$$

$$|y| > 0$$

Now, pumping y ,

$|xy| \leq p \rightarrow xy$ consists of only for a 's.

$$xyy'z = a^{(p+m)}$$

Here, it is not in L_4 ;

as the number of a 's might not ^{be} a prime number here; for every case.

$\therefore$ There is a contradiction.

Step-4: Conclusion:

Therefore, L_4 is non-regular.

[proved]

[P.T.O.]

Answer to the Q. NO - 01(e)

Given,

$$L_5 = \{ w \in \{0\}^* : 0^{3^n} \text{ where } n \geq 0 \}$$

Step - 1: Assumption:

let us assume,
 L_5 is regular.

Step - 2: Pumping Length:

let,
 p = pumping length.

Step - 3: Choose:

for $s_5 = 0^{3^n} \in L_5$ where,
Now, $= 0^{3p} \in L_5$ $3^n \geq p; n \geq 0$
Splitting s_5 into xyz ; it satisfies;
 $|s| = 3^n$.

$s'_5 = 0^x 0^y 0^z$ where,
here, $|xy| \leq p$
 $|y| > 0$
 $y = 0^k$
(since y consist only of 0s)
 $x = p - k$
 $y = k$
 $z = 2p$; &
 $0 < k \leq p$

Now, pumping y ,

$$\begin{aligned} xy^kz &= 0^{(p+k)} 0^k 0^{2p} \\ &= 0^{(p+k)} 0^{(2p)} \\ &= 0^{(3p+k)} \end{aligned}$$

Here, this pumping adds $k \rightarrow 0$ s into the
[P.T.O.]

String ;

$\therefore x > 0$ & p is constant;

Hence, the length of the x new pumped string will not always be a multiple of 3.

$\therefore$ There is a contradiction.

Step-4: Conclusion:

Therefore, L_5 is non-regular.

[proved]

There are a total of three problems. You have to solve all of them!

Problem 1 (CO3): Context-Free Grammars (10 points)

Let $\Sigma = \{a, b\}$. Consider the following languages over Σ . Recall that for a string w , $|w|$ denotes the length of w .

$$L_1 = \{w \mid w \text{ is an even length palindrome}\}$$

$$L_2 = \{w \mid \text{Length of } w \text{ is even}\}$$

$$L_3 = \{x11y \mid x, y \in L_2, |x| = |y|\}$$

$$L_4 = L_1 \cap L_3$$

- (a) **Give** a context free grammar for L_1 . (3 points)
- (b) **Give** a context free grammar for L_3 . (4 points)
- (c) **Give** a context free grammar for L_4 . (3 points)

Problem 2 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Take a look at the grammar below and solve the following problems.

$$A \rightarrow 1A \mid 1C \mid 0B \mid 00A$$

$$B \rightarrow 0A \mid 1B \mid 00B$$

$$C \rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid \epsilon$$

- (a) **Give** a leftmost derivation for the string 01011001. (3 points)
- (b) **Sketch** the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) **Demonstrate** that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string. (3 points)
- (d) **Find** a string w of length six such that w has exactly one parse tree in the grammar above. (2 point)

Problem 3 (CO4): Pushdown Automata (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages.

$$L_1 = \{w \mid w \text{ starts and ends with the same character}\}$$

$$L_2 = \{w \mid \text{the number of 0s in } w \text{ is not the same as the number of 1s}\}$$

- (a) **Give** the state diagram of a pushdown automaton that recognizes L_1 . (4 points)
- (b) **Give** the state diagram of a pushdown automaton that recognizes L_2 . (6 points)

Assignment - 03

Submitted by : TASNIM RAHMAN MOUMITA

ID : 22301689

Course Title : Automata & Computability

Course Code : CSE331

Section : 09

Date of submission : 08.01.2025

Semester : Fall 2024

There are a total of three problems. You have to solve all of them!

Problem 1 (CO3): Context-Free Grammars (10 points)

Let $\Sigma = \{a, b\}$. Consider the following languages over Σ . Recall that for a string w , $|w|$ denotes the length of w .

$$L_1 = \{w \mid w \text{ is an even length palindrome}\}$$

$$L_2 = \{w \mid \text{Length of } w \text{ is even}\}$$

$$L_3 = \{x11y \mid x, y \in L_2, |x| = |y|\}$$

$$L_4 = L_1 \cap L_3$$

- (a) **Give** a context free grammar for L_1 . (3 points)
- (b) **Give** a context free grammar for L_3 . (4 points)
- (c) **Give** a context free grammar for L_4 . (3 points)

Problem 2 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Take a look at the grammar below and solve the following problems.

$$A \rightarrow 1A \mid 1C \mid 0B \mid 00A$$

$$B \rightarrow 0A \mid 1B \mid 00B$$

$$C \rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid \epsilon$$

- (a) **Give** a leftmost derivation for the string 01011001. (3 points)
- (b) **Sketch** the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) **Demonstrate** that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string. (3 points)
- (d) **Find** a string w of length six such that w has exactly one parse tree in the grammar above. (2 point)

Problem 3 (CO4): Pushdown Automata (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages.

$$L_1 = \{w \mid w \text{ starts and ends with the same character}\}$$

$$L_2 = \{w \mid \text{the number of 0s in } w \text{ is not the same as the number of 1s}\}$$

- (a) **Give** the state diagram of a pushdown automaton that recognizes L_1 . (4 points)
- (b) **Give** the state diagram of a pushdown automaton that recognizes L_2 . (6 points)

Q Context - Free Grammars :

Answer to the Q. NO- 01 (a)

Given,

$$L_1 = \{w \text{ is an even length palindrome}\}$$

$$L_2 = \{\text{Length of } w \text{ is even}\}$$

$$L_3 = \{x \parallel y \mid x, y \in L_2, |x| = |y|\}$$

$$L_4 = L_1 \cap L_3$$

for L_1 ,

an even length palindrome:-

for $w = \epsilon : \epsilon \in L_1$; generated by $S \rightarrow \epsilon$

for $w = aa : S \rightarrow aSa \rightarrow aa$

for $w = abba : S \rightarrow aSa \rightarrow abSba \rightarrow abba$

$\therefore$ context - free grammar for L_1 ;

$$\boxed{S \rightarrow aSa \mid bSb \mid \epsilon}$$

here,

ϵ represents
empty string, which
ensures base case.

(Ans)

[P.T.O.]

Answer to the Q. NO-01(b)

Given,

$$L_3 = \{ x11y \mid x, y \in L_2, |x| = |y| \}$$

$$L_2 = \{ \text{length of } w \text{ is even} \}$$

let us define,

S_3 for L_3

and S_2 for L_2

context free grammar formation:-

$$\text{for } S_3 \rightarrow S_2 11 S_2$$

$$\text{for } S_2 \rightarrow a S_2 a \mid b S_2 b \mid \epsilon$$

Which verifies:-

(Ans:-)

$$S_2 \rightarrow aa$$

$$S_2 \rightarrow abba, \text{ etc.}$$

$$S_3 \rightarrow S_2 11 S_2$$

[This ensures x, y are generated with equal lengths]

$$\text{for } x = \epsilon,$$

$$y = \epsilon,$$

$$S_3 \rightarrow \epsilon 11 \epsilon = 11$$

$$\text{for } x = aa,$$

$$y = bb$$

$$S_3 \Rightarrow S_2 11 S_2 \rightarrow aa 11 bb$$

(Ans:-)

[P.T.O.]

Answers to the Q. NO-01(c)

Given,

$$L_1 = \{w \text{ is even length palindrome}\}$$

$$L_3 = \{x11y \mid x, y \in L_2, |x| = |y|\}$$

$$L_4 = L_1 \cap L_3$$

Context free Grammar for L_4 :-

$$S \rightarrow aSa \mid bSb \quad \left[\begin{array}{l} \text{Generates even length} \\ \text{palindrome} \end{array} \right]$$

$$M \rightarrow aM \mid bM \mid \epsilon \quad \left[\begin{array}{l} \text{Generates even length} \\ \text{strings} \end{array} \right]$$

$$S' \rightarrow M11M \quad \left[\begin{array}{l} \text{Generates strings of the} \\ \text{form } x11y; \text{ that ensures} \\ x \text{ and } y \text{ are even length} \end{array} \right]$$

$\therefore$ The context free Grammar for L_4 :

$$\boxed{\begin{array}{l} S \rightarrow aSa \mid bSb \mid M11M \\ S' \rightarrow aM \mid bM \mid \epsilon \end{array}}$$

(Ans)

[P.T.O.]

Answer to the Q. NO - 02 (a)

Q Derivations, Parse Trees and Ambiguity:

Given,

$$A \rightarrow 1A \mid 1C \mid 0B \mid 00A$$

$$B \rightarrow 0A \mid 1B \mid 00B$$

$$C \rightarrow 0C0 \mid 0C1 \mid 1C1 \mid \epsilon$$

$$\text{String} \rightarrow 01011001.$$

Now,

leftmost derivation:-

$$\overset{A}{\Rightarrow} 0B$$

$$\Rightarrow 00A$$

$$\Rightarrow 001C$$

$$\Rightarrow 0011C0$$

$$\Rightarrow 0011001$$

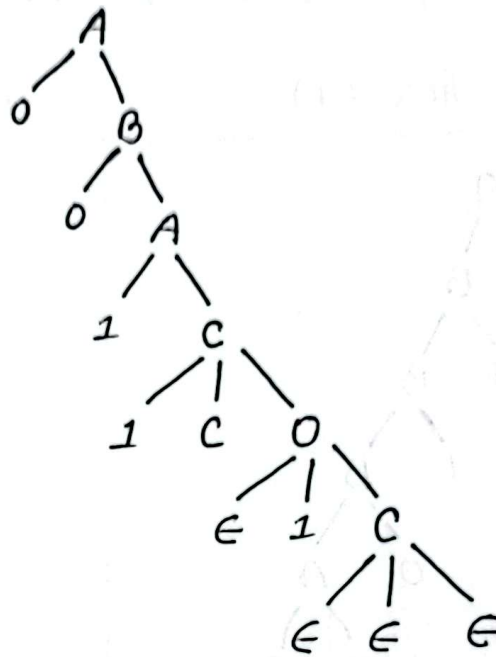
(Ans)

[P.T.O.]

Answer to the Q. NO- 02(b)

For the leftmost derivation I got in answer of 02(a),

The Parse Tree is drawn below:-



(Ans)

[p.t.o.]

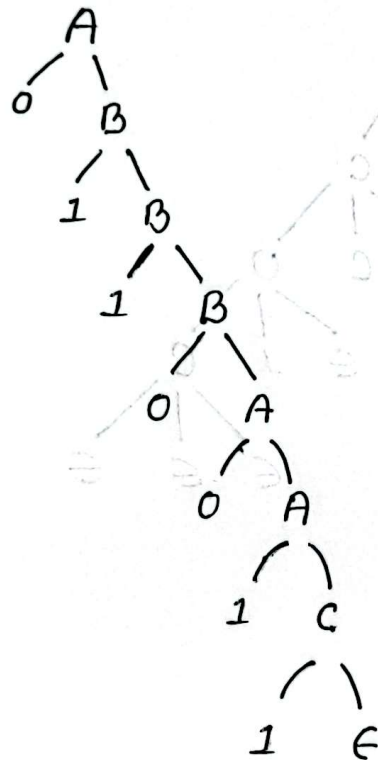
Answer to the Q. NO - 02(c)

Ambiguity demonstration:

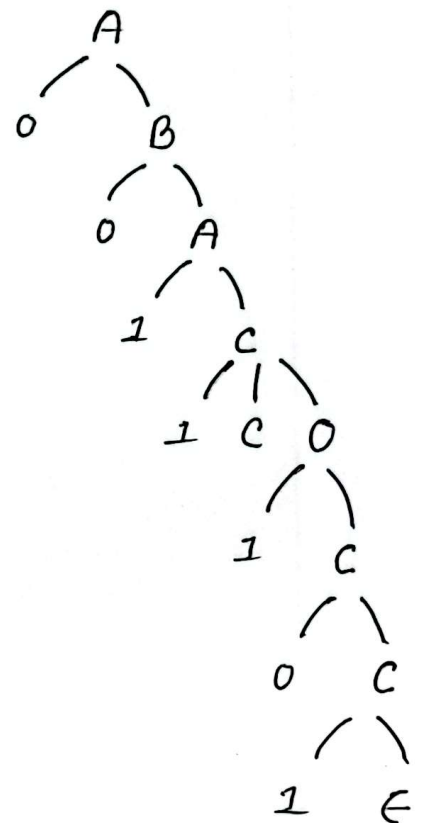
except 0101101, [found in 2(b)]

Two more parse trees are drawn below:

Parse Tree (i)



Parse Tree (ii)



(Ans)

[P.T.O.]

Answer to the Q. NO-02(d)

Given grammar,

$$A \rightarrow 1A \mid 1C \mid 0B \mid 00A$$

$$B \rightarrow 0A \mid 1B \mid 00B$$

$$C \rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid \epsilon$$

To find,

a string of length six such that w has exactly one parse tree in the given grammar.

let,

The string of length six = 000000

$\therefore$ leftmost derivation:

$$\begin{array}{l} A \\ \Rightarrow 00A \end{array}$$

$$\Rightarrow 0000A$$

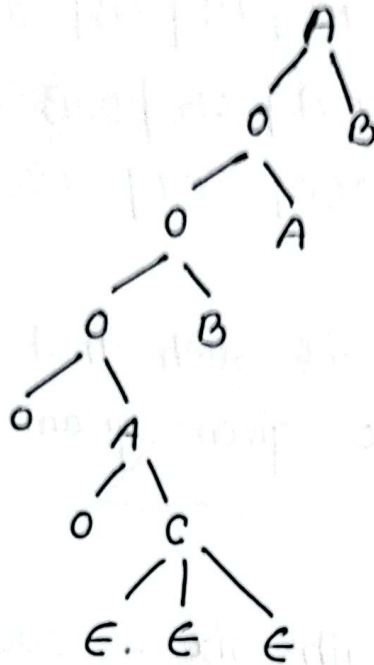
$$\Rightarrow 000000A$$

$$\Rightarrow 000000 \quad \left[\text{using rule, } A \rightarrow \epsilon \right]$$

This string '000000' of six length has only one possible parse tree.

[The parse tree is drawn in next page.] [P.T.O.]

parse tree for string : 000000



(Ans)

Pushdown Automata :

Answer to the Q. NO - 03 (a)

Given,

let

$$\Sigma = \{0, 1\}$$

$$L_1 = \{w \mid w \text{ starts and ends with same character}\}$$

here,

symbol, pop $\rightarrow$ push

State diagram:-

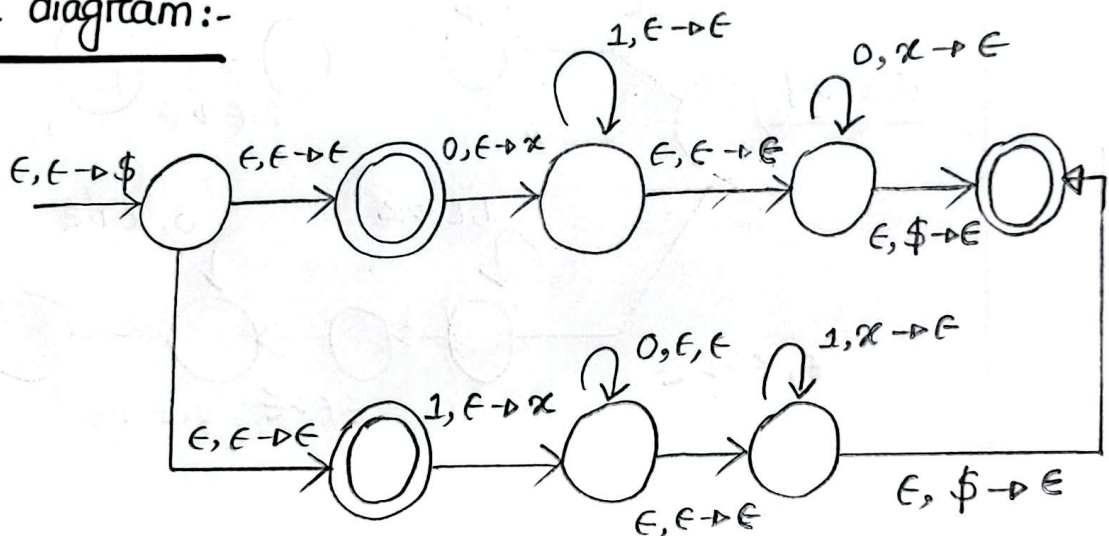


Figure: PDA state diagram for L_1 .

(Ans)

[P.T.O.]

Answer to the Q. NO - 03(b)

Given,

let,

$$\Sigma = \{0, 1\}$$

$$L_2 = \{w \mid \text{the number of 0s in } w \text{ is not the same as the number of 1s}\}$$

State diagram:

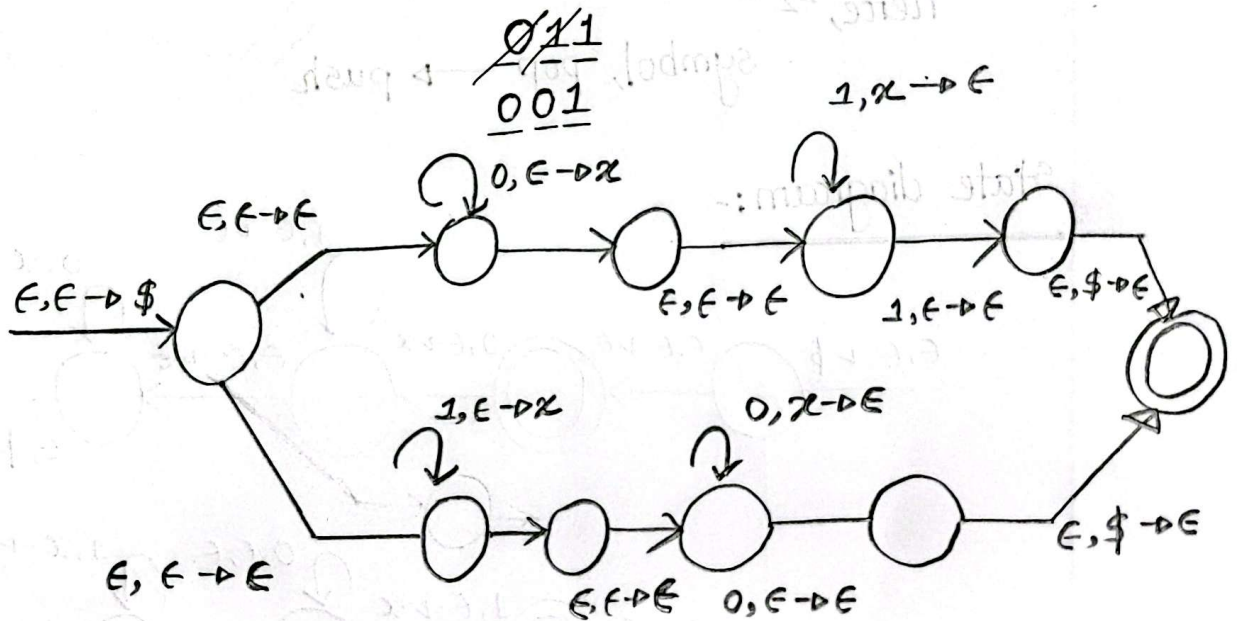


Figure : PDA State Diagram for L_2 .

(Ans:)